Nexora – Zstandard HTTP Response Compression

Technical Performance Evaluation

Component: Django Zstandard Response Middleware
Technology: Python, Django, Zstandard
Tested Compression Levels: 3, 11 and 22
Status: Initial testing completed

1. Objective

The objective of this implementation is to reduce HTTP response sizes generated by Django applications through Zstandard compression.

The middleware aims to:

- Reduce response data size.
- Reduce network data transfer.
- Measure compression processing time.
- Monitor CPU consumption.
- Measure Python memory allocations.
- Evaluate compression efficiency across different levels.

2. Testing Methodology

Three Zstandard compression levels were evaluated:

- Level 3
- Level 11
- Level 22

Tests were performed on actual Django application responses, including the homepage and Django Admin pages.

Measurements included original response size, compressed response size, saved bytes, compression reduction percentage, wall-clock time and CPU time.

3. Compression Results

Page	Original	Level 3	Level 11	Level 22
Homepage	840,000 B	136 B	136 B	130 B
Admin	21,819 B	3,129 B	2,885 B	2,796 B
Admin Home	12,650 B	2,391 B	2,286 B	2,227 B
Admin Login	4,084 B	1,578 B	1,523 B	1,497 B

4. CPU Performance Results

Latest recorded measurements:

Page	Level 11	Level 22
Homepage	5.538 ms	6.615 ms
Admin	0.529 ms	11.939 ms
Admin Home	0.698 ms	6.598 ms
Admin Login	0.674 ms	0.919 ms

These are individual measurements, not averages. Repeated controlled tests are required for reliable performance comparisons.

5. Key Findings

Finding 1 – Response Size Reduction

All three compression levels successfully reduced the tested response sizes.

Finding 2 – Compression Level Comparison

Level 22 produced the smallest compressed output. However, its additional size reduction over level 11 was relatively small on the tested Admin pages.

Finding 3 – CPU Consumption

The latest measurements showed substantially higher CPU time for level 22 on the Admin and Admin Home pages compared with level 11.

Finding 4 – Homepage Compression

The homepage response was compressed from 840,000 bytes to 130–136 bytes. This unusually high reduction indicates highly repetitive content and should not be treated as representative of ordinary production website data.

6. Current Recommendation

Recommended candidate: Zstandard Level 11

Level 11 currently appears to provide a promising balance between compression efficiency and CPU consumption.

This is a provisional recommendation. More repeated testing is necessary before selecting a final production configuration.

7. Limitations

- Current CPU measurements are individual observations.
- Memory measurements require further validation.
- The homepage contains unusually compressible content.
- Actual internet transfer performance has not yet been established.
- Streaming responses are not currently supported by this middleware.
- Real-world performance under concurrent requests has not been evaluated.

8. Future Improvements

1. Perform at least 10 compression runs per level.
2. Calculate average and median compression time.
3. Improve memory measurement accuracy.
4. Benchmark realistic HTML, JSON and API responses.
5. Compare total HTTP response times.
6. Evaluate performance under concurrent requests.
7. Investigate streaming response compression.
8. Evaluate production deployment behind a reverse proxy.

9. Conclusion

The initial evaluation confirms that the Nexora Zstandard middleware successfully compresses eligible Django HTTP responses.

The results demonstrate substantial response-size reduction across the tested pages.

Further performance testing is required to determine the optimal compression configuration based on response size, CPU consumption, memory usage and overall response time.

Project: Nexora
Component: HTTP Response Optimization

Implementation: Zstandard Response Middleware
Evaluation Status: Initial benchmark completed